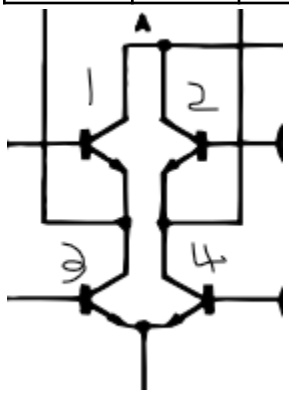
CSEN/ECEN 123 Lab 4 Prelab

Austin Petersen, Dylan Thornburg, Madeline Follosco

Study the H-bridge. For each combination of inputs listed below, provide the voltage outputs at Out1 and Out2 that drive the motor and describe which transistors are on and which are off. (See the data sheet and manual). Assume a 9.6V battery. Also, explain which behavior will be seen for each input combination (forward, backwards, brake (fast stop), or coast (free run)) and explain why that behavior will occur.

EnA	In1	In2	Out1	Out2	Behavior
L	L	L	High Z, no transistors on	High Z, no transistors on	Coast
L	L	H	High Z, no transistors on	High Z, no transistors on	Coast
L	H	L	High Z, no transistors on	High Z, no transistors on	Coast
L	H	H	High Z, no transistors on	High Z, no transistors on	Coast
H	L	L	0 | 0V, transistor 3 on, transistor 1 off	0 | 0V, transistor 4 on, transistor 2 off	Brake
H	L	H	0 | 0V, transistor 3 on, transistor 1 off	1 | 9.6V, transistor 2 on, transistor 4 off	Reverse
H	H	L	1 | 9.6V, transistor 1 on, transistor 3 off	0 | 0V, transistor 4 on, transistor 2 off	Forward
H	H	H	1 | 9.6 V, transistor 1 on, transistor 3 off	1 | 9.6 V, transistor 2 on, transistor 4 off	Brake



• Consider the case where you are going to change the motor from running forwards to backwards. You cannot change all 3 motor control signals at once. As you change one at a time, you will enter intermediate states. Figure out what possible intermediate states you could enter and list them. Write an algorithm to switch from forwards to backwards motor operation that passes through the least disruptive intermediate states.

Well it depends on if you want to brake or not.

If you don't want to brake to slow before switching:

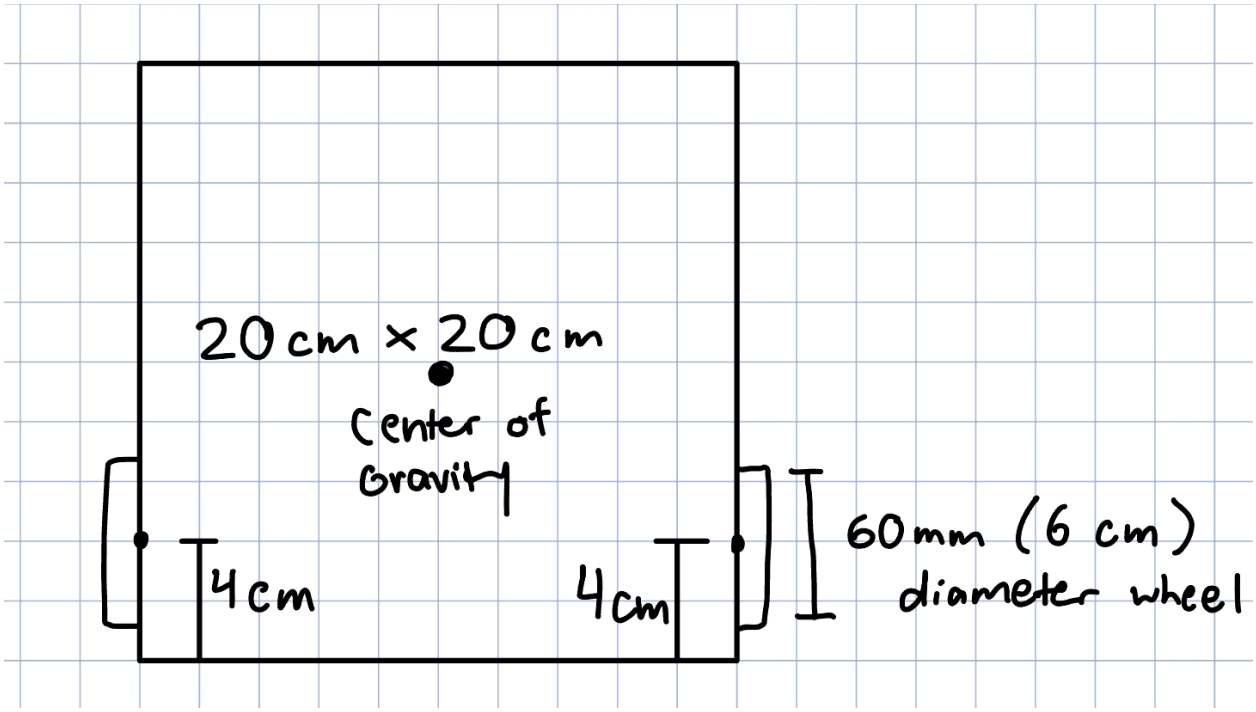
EnA	In1	In2	Behavior
H	H	L	Forward
L	H	L	Coast
L	L	L	Coast
L	L	H	Coast
H	L	H	Backward

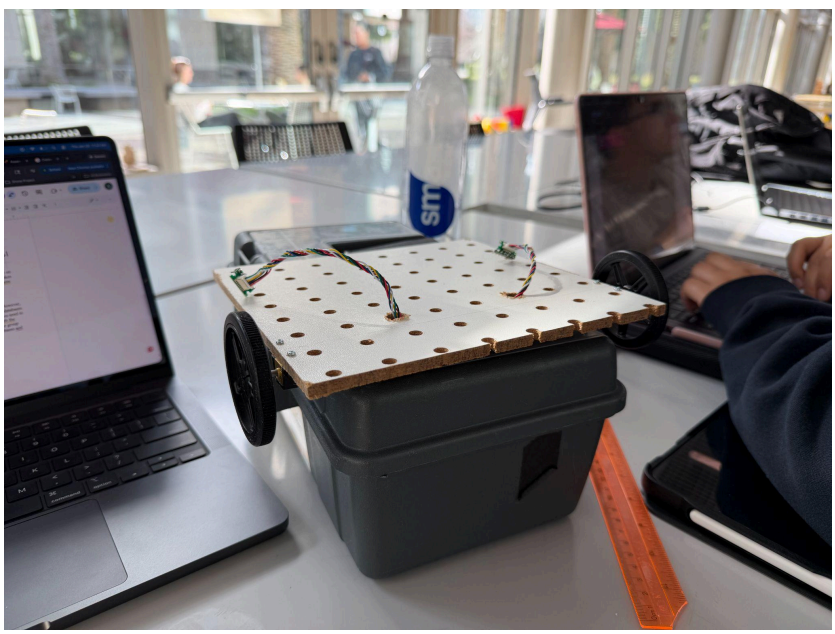
If you DO want to brake to slow before switching:

EnA	In1	In2	Behavior
H	H	L	Forward
H	L	L	Brake
H	L	H	Reverse

Even though the first algorithm has more steps, I'd argue it's less disruptive and therefore closer to what the problem is asking for.

• Document the length, width, and wheel position for your platform and explain why you chose them. Provide a hand-drawn or CAD sketch of your platform.





We chose to place the 2 wheels in the back of the frame for more precise steering. They were on the outer edges of the frame to provide stability and a wider distance between the wheels makes turning in place easier. Both wheels were placed evenly from the center of gravity of the frame for balance and straight line performance.

Board dimensions: 20 cm x 20 cm ish (one side is 20.8 cm)

• Did you know that the University has a library? Rumor is that they have books there. However, nobody ever reads those books anymore. The library also has access to many scholarly databases which contain interesting facts, some of which are so old that they pre-date Tik-Tok. You need to

download an ancient article from the IEEE Xplore database. Locate my 1994 paper with the subtitle “A First Step Towards Software Power Minimization.” Include a selfie of your group with the first page from that paper. The library has instructions on how to access databases and librarians can help.